Best Practices for Biomedical Research Data Management

Module 7: Data Sharing and Reuse
Open Data & Data Sharing

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This video will expand on open data and benefits of data sharing. Data sharing builds on efforts taken earlier in the research data life cycle, and this video will describe how good data management throughout the research project can help researchers share their data more effectively, and also how open data leads to more citations, collaborations and reproducibility.

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Open data in science

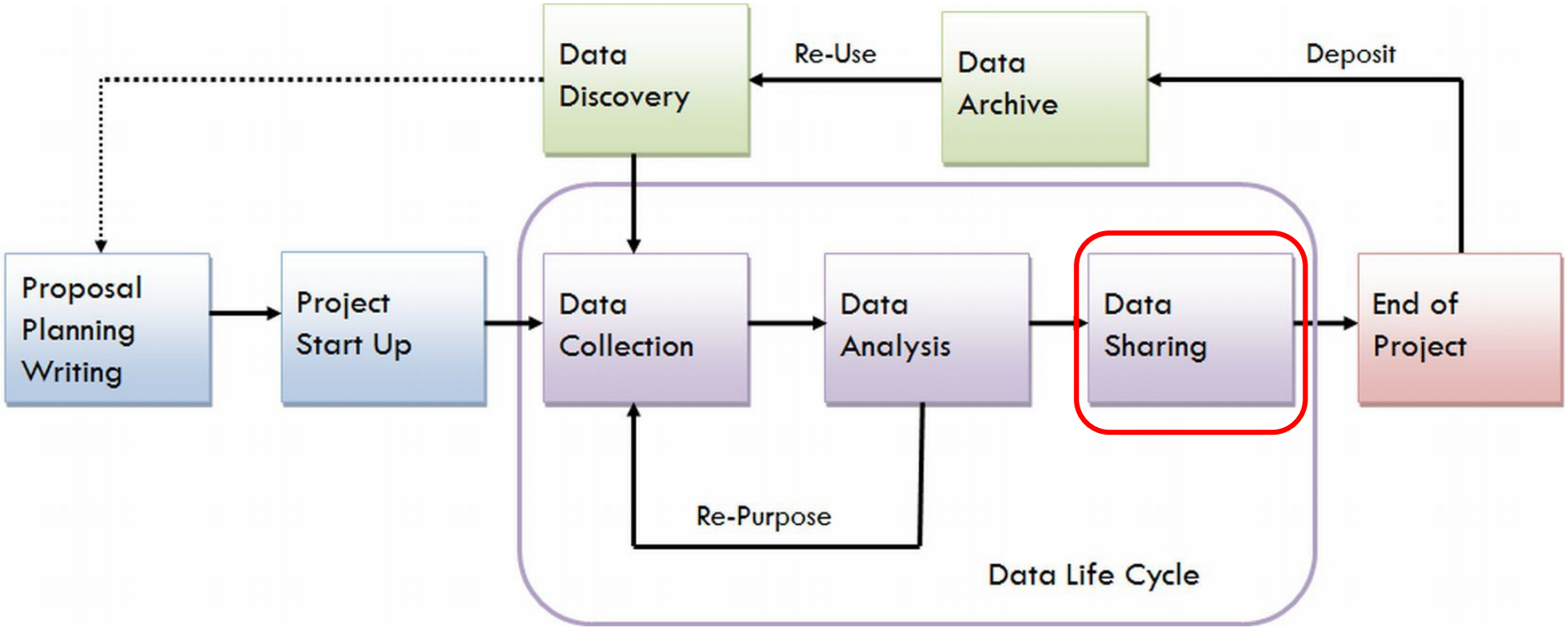
"By open data in science we mean that it is freely available on the public internet permitting any user to download, copy, analyse, re-process, pass them to software or use them for any other purpose without financial, legal, or technical barriers other than those inseparable from gaining access to the internet itself. To this end data related to published science should be explicitly placed in the public domain."

Murray-Rust et al. (2010) Panton Principles for open data in science.
http://pantonprinciples.org/

Data sharing

The practice of making research data available to others for validation and replication of results.

Research data life cycle



http://data.library.virginia.edu/data-management/lifecycle/

Why share data?

The sticks…

- ➔ Funder requirements
- ➔ Publisher requirements
- ➔ Freedom of Information Act

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Why share data?

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The carrots…

- ➔ Increases the impact and visibility of research
- ➔ Provides credit to the researcher as a research output in its own right
- ➔ Enables new scientific collaborations
- ➔ Reduces duplication of effort
- ➔ Maximizes transparency, accountability and scrutiny of research findings

Impact and visibility of research

Published articles with shared data sets receive more citations than articles without shared data.

PeerJ

✔ PEER-REVIEWED

Data reuse and the open data citation advantage

Bioinformatics | Science Policy

Heather A. Piwowar[1,2], Todd J. Vision[1,2,3]

Published October 1, 2013

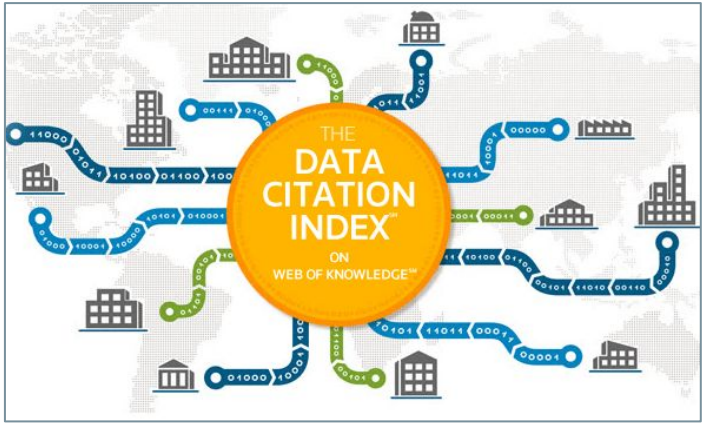
*"**Conclusion.** After accounting for other factors affecting citation rate, we find a robust citation benefit from open data, although a smaller one than previously reported. We conclude there is a direct effect of third-party data reuse that persists for years beyond the time when researchers have published most of the papers reusing their own data."*

Credit to the researcher

Data can be cited in the same way that articles are cited

Data citations can be tracked in the same way that articles are tracked

Data citations can be included in CVs and biosketches



How to give credit

Recommended format for a data citation:

Creator (Publication Year). Title. Publisher. Identifier.

Example:

Prayle AP, Hurley MN, Smyth AR (2012) Data from: Compliance with mandatory reporting of clinical trial results on ClinicalTrials.gov: cross sectional study. Dryad Digital Repository. http://dx.doi.org/10.5061/dryad.j512f21p

Collaboration and effort

Data can be reused or recombined in unexpected ways

Data can be used to plan new studies

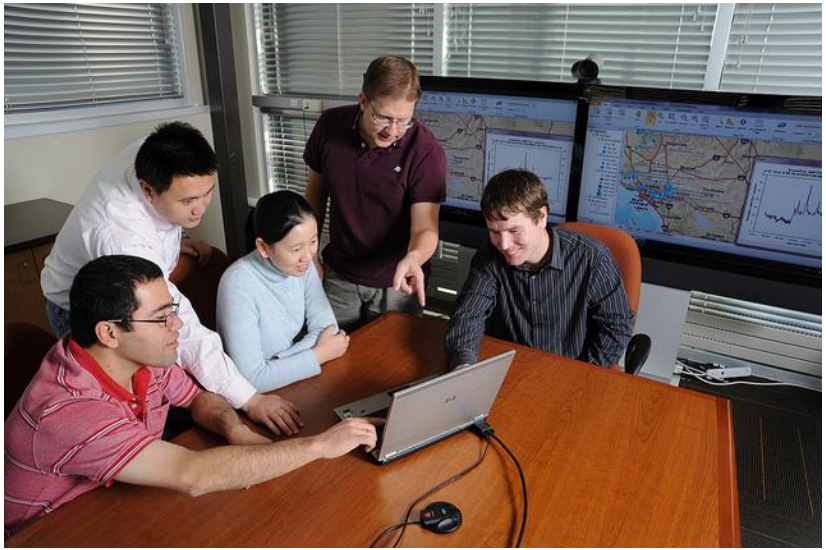
Produce better data documentation

Focus resources and efforts appropriately

Avoid repeating studies

Avoid repeating mistakes

Work smarter and more quickly



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Transparency and reproducibility

Retraction Watch

Tracking retractions as

Diederik Stapel now has 58 retractions

with 4 comments

Social psychologist Diederik Stapel has notched his 58th retraction, after admitting he fabricated data in yet another article.

He's holding onto his 4th place spot on our leaderboard.

This latest retraction is for "Correction or comparison? The effects of prime awareness on social judgments," published in the *European Journal of Social Psychology*. As usual for Stapel, this paper has been retracted because he fabricated data.

Here's the note:

> The above article from the *European Journal of Social Psychology*, published online on 17 February 2009 in Wiley Online Library (wileyonlinelibrary.com), and in

Meet the 28-Year-Old Grad Student Who Just Shook the Global Austerity Movement

PLOS data policy

"PLOS journals require authors to make all data underlying the findings described in their manuscript fully available without restriction, with rare exception."

Effective March 1, 2014
http://journals.plos.org/plosone/s/data-availability

Sharing data publicly

Data cleaning, quality control

Data documentation

Data deposition

- Minimally, data that supports a publication
- Data cleaning and documentation are critical
- Deposit to a repository

How to choose a repository

Does the journal make any recommendations for deposit?

Does your discipline use a specific data repository?

Are there local repositories at your disposal, e.g. an institutional repository?

Consult a registry (re3data.org or OAD)

Recommended data repositories

PLOS
http://journals.plos.org/plosone/s/data-availability#loc-recommended-repositories

Nature Publishing Group
http://www.nature.com/sdata/policies/repositories

Permanent, unique identifiers

When choosing a repository, look for one that assigns a permanent unique identifier to your data

For example: a DOI, or Digital Object Identifier

Structure of a DOI: 10.7191/jgr.2016.1021

DOI directory | Publisher | Unique identifier assigned by publisher

Disciplinary repositories

ICPSR

GEO
Gene Expression Omnibus

FlyBase

NCBI
GenBank

DataONE

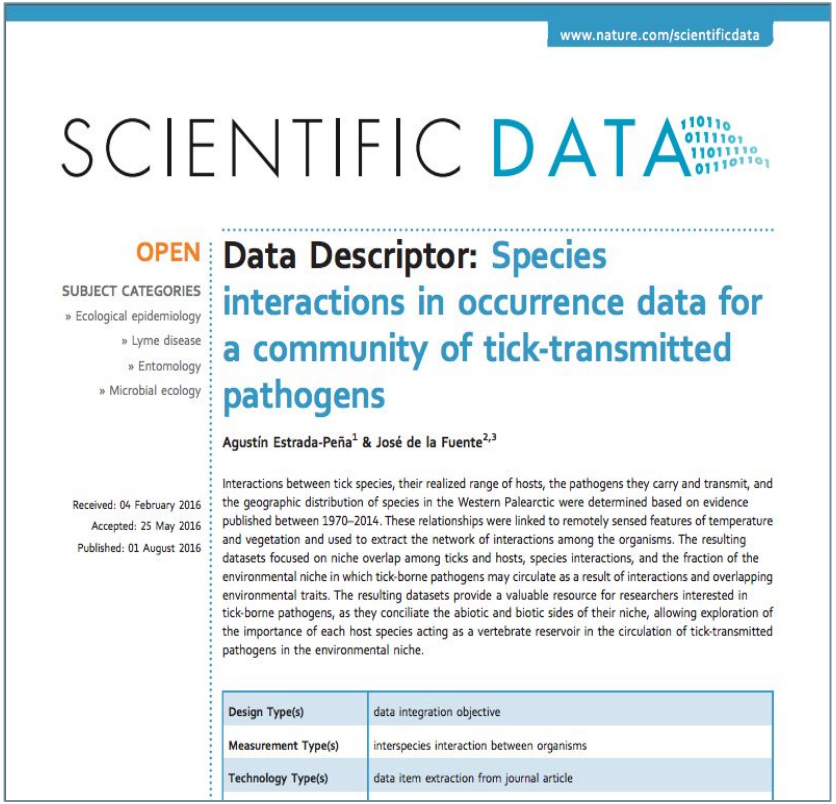
General use / multidisciplinary repositories

Institutional repositories

Data journals

SCIENTIFIC DATA

(GIGA)n SCIENCE



www.nature.com/scientificdata

SCIENTIFIC DATA

OPEN

SUBJECT CATEGORIES
» Ecological epidemiology
» Lyme disease
» Entomology
» Microbial ecology

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Data Descriptor: Species interactions in occurrence data for a community of tick-transmitted pathogens

Agustín Estrada-Peña[1] & José de la Fuente[2,3]

Interactions between tick species, their realized range of hosts, the pathogens they carry and transmit, and the geographic distribution of species in the Western Palearctic were determined based on evidence published between 1970–2014. These relationships were linked to remotely sensed features of temperature and vegetation and used to extract the network of interactions among the organisms. The resulting datasets focused on niche overlap among ticks and hosts, species interactions, and the fraction of the environmental niche in which tick-borne pathogens may circulate as a result of interactions and overlapping environmental traits. The resulting datasets provide a valuable resource for researchers interested in tick-borne pathogens, as they conciliate the abiotic and biotic sides of their niche, allowing exploration of the importance of each host species acting as a vertebrate reservoir in the circulation of tick-transmitted pathogens in the environmental niche.

Design Type(s)	data integration objective
Measurement Type(s)	interspecies interaction between organisms
Technology Type(s)	data item extraction from journal article

List of data journals
http://proj.badc.rl.ac.uk/preparde/blog/DataJournalsList